i. Alpha herpes viruses

They have relatively **short replicative cycle** (12–18 hours) and a variable host range. They cause latent infection in sensory ganglia. Example: Herpes simplex virus and varicella zoster virus.

ii. Beta herpes viruses

They **replicate slowly** (more than 24 hours) and have a narrow host range, grow well in fibroblasts. They cause latent infection of salivary gland and other organs. Example: Cytomegalovirus.

iii. Gamma herpes viruses

They have a **narrow host range** and replicate in lymphoblastoid cells. They are specific for either B or T lymphocytes and causes latent infection in lymphoid tissue Example: Epstein - Barr Virus. Eight different types of herpes viruses are known whose primary hosts are humans. They have been designated as 'Human herpes virus type 1–8.

1. Herpes Simplex

The herpes simplex virus (HSV) occurs **naturally only in humans**, but it can produce experimental infection in laboratory animals. There are **two types** of the herpes simplex virus. **HSV type 1** (Human herpes virus type 1 or **HHV type 1**) is isolated from lesions in and around the mouth and is transmitted by direct contact or droplet spread from carrier. **HSV type 2** (Human herpes virus type 2 or **HHV type 2**) is responsible for the genital herpes infections transmitted venereally.

Pathogenesis

Herpes simplex is one of the most **common viral infection** humans, the sources of infection are saliva, skin lesions or respiratory secretions. In type 2, transmission occurs by close contact and may be venereal in genital herpes.

The virus enters through defects in the skin or mucous membranes and multiples locally, with cell to cell spread. The **herpes lesions** are thin walled, umbilicated vesicles, the roof of which breaks down, leaving tiny superficial ulcers. They heal without scarring.

Clinical features

The clinical manifestations depend on the site of infection, age and immune status of the host and the antigenic type of the virus. They are

- Cutaneous infections
- Mucosal infections
- Ophthalmic infections
- Nervous system infections
- Visceral infections
- Genital infections

Laboratory diagnosis

Microscopy

Smears are prepared from the lesions, from the vesicles and stained with 1% aqueous solution of **toluidine blue 'O'** for 15 seconds. Multinucleated giant cells with faceted nuclei with ground glass chromatin (**Tzanck cells**) was observed.

Virus isolation

Inoculation in mice and on chick embryo CAM is insensitive. Primary human embryonic kidney, human amnion cells are susceptible, but **human diploid fibroblasts** are preferred vesicle fluid, spinal fluid, saliva and swabs may be used. Cytopathic changes may appear as early as 24–48 hrs.

TN_GOVT_XII_Micro_Biology_CH10.indd 213 04-03-2019 12:27:07